

TerraState: A Testable Predictive-State World Model for Weather-Driven Land-Surface Forecasting

Anonymous submission

Abstract

High-resolution satellite time series are a primary tool for monitoring vegetation, agriculture, and ecosystem response, and are increasingly cast as weather-driven forecasting: predicting future land-surface observations from cloud-obscured image histories and meteorological drivers. Yet such models are selected almost entirely by fixed-horizon pixel accuracy, which cannot establish whether their internal representations act as reusable world states rather than one-shot predictors. An accurate forecaster may still ignore the weather forcing, collapse toward persistence, or reach mutually incompatible states when the same interval is traversed directly versus in steps—failures that standard error metrics cannot detect. We introduce **TerraState**, a testable predictive-state world model that augments a forecasting backbone with a spatial state inferred from cloud-masked histories and a shared transition conditioned on future weather, geography, and elapsed time, requiring this state to carry the forecast. Rather than asserting a world state by architecture, TerraState makes it falsifiable through matched direct-versus-recursive forecasts, shuffled- and zeroed-driver controls, and direct-versus-composed consistency, each guarded by endpoint accuracy. On GreenEarthNet under temporal distribution shift, we report forecast skill—against persistence, video-prediction, and vegetation-forecasting baselines and a matched-backbone, matched-budget ablation—separately from these state diagnostics, turning the claim that a forecaster has learned a world state into a falsifiable question: whether an accurate forecaster also exposes a non-degenerate, driver-sensitive, and composition-consistent predictive state.

1 Introduction

High-resolution Earth-observation (EO) time series make localized vegetation dynamics directly observable, but only intermittently: optical acquisitions are sparse and often obscured by clouds, while the relevant land-surface response depends on weather and static geography. EarthNet2021 formalized this setting as guided video prediction from past Sentinel-2 observations, topography, and future meteorology (Requena-Mesa et al. 2021). GreenEarthNet subsequently introduced a vegetation-focused cloud mask, a temporal-

distribution-shift evaluation, and Contextformer, which combines a spatial vision backbone with weather-conditioned temporal prediction at 20 m resolution (Benson et al. 2024). Recurrent, convolutional, transformer, and diffusion models now provide strong alternatives for spatiotemporal forecasting (Shi et al. 2015; Wang et al. 2017; Gao et al. 2022b,a; Voleti, Jolicoeur-Martineau, and Pal 2022). Recent EO-specific work further models predictive uncertainty (Zhao et al. 2025), structures meteorological forcing and evaluates response fidelity (Luo et al. 2026), or rolls a latent vegetation state under user-specified weather scenarios (Iele et al. 2026).

These advances sharpen a question that fixed-horizon forecast error does not answer: *when does an accurate forecaster expose a state that can actually be advanced and reused?* A low pixel error can coexist with a representation that mainly copies the last clear observation, reacts weakly to future weather, or follows incompatible trajectories when the same forcing interval is traversed in one step or in several steps. General time-series evidence similarly shows that accurate forecasts can be supported by temporally disordered latent representations (Yang et al. 2026). In such cases, calling an internal feature a “world state” describes an architecture, not an empirically supported capability.

We argue that a predictive-state claim should be operational and falsifiable. At minimum, the claimed state must contribute measurably to an accurate endpoint forecast, and its shared transition must respond to the declared exogenous driver. A stricter, optional test asks whether different partitions of the same driver path reach compatible, accurate endpoints without identity dynamics or representation collapse. These requirements are narrower than claiming a complete physical state or a causal simulator. They are also complementary to the output-level weather-response tests of Luo et al. (2026) and the scenario-conditioned latent rollout of Iele et al. (2026): those works ask whether outputs respond plausibly to forcing, whereas we ask whether an exposed spatial state is load-bearing and whether its shared transition withstands matched internal and endpoint tests.

TerraState implements this view with three components. A context operator combines cloud-masked EO history, retained past meteorology, and static geography to infer a spatial state; a single transition advances that state under the 24-channel future-weather path, the same static geography, and forecast horizon; and a forecast closure makes the

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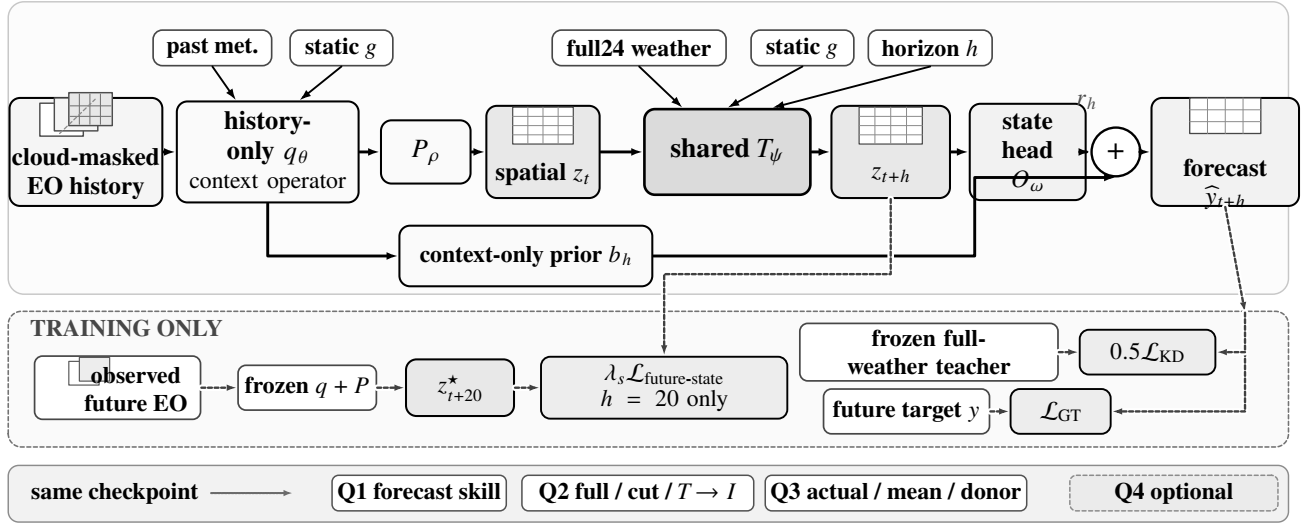


Figure 1: TerraState inference and training supervision. Cloud-masked EO history, retained past meteorology, and static geography enter the history-only context operator q_θ , which emits a context-only prior b_h and, through P_ρ , a spatial predictive state z_t . Full24 future weather, static geography, and horizon h condition the shared transition T_ψ ; O_ω decodes z_{t+h} into a contribution r_h that is added explicitly to b_h . Solid arrows denote inference. Dashed branches are training only: ground-truth forecasting, distillation from a frozen full-weather teacher, and an $h = 20$ future-state anchor produced from observed future EO by a frozen target encoder. Neither teacher nor future observation is used at inference. The narrow bottom strip is a space-constrained index of same-checkpoint verification; it does not depict an additional model component.

transitioned state carry an identifiable contribution beyond a context-only prior. Training aligns the terminal transitioned state to a frozen representation of the observed future. We separate ordinary forecasting skill (Q1) from load-bearing cuts (Q2) and matched driver controls (Q3); direct-versus-composed path consistency and anti-collapse measurements form an optional post-training extension (Q4).

Our contributions are:

- We operationalize a predictive-state world model for weather-driven, high-resolution land-surface forecasting: a context-derived spatial state is advanced by one full24-conditioned shared transition and contributes to the endpoint forecast beyond a context-only prior.
- We anchor the transitioned state to a representation of the observed future produced by a frozen target encoder, connecting future-observation supervision to a state that remains on the forecast path at inference.
- We report forecast skill and state evidence separately for one selected checkpoint. Q1–Q3 test forecast sufficiency, load-bearing state use, and weather response; optional Q4 probes direct/composed reuse under endpoint and non-collapse guards. This is a matched verification protocol, not a new benchmark.

2 Related Work

Weather-driven EO forecasting. EarthNet2021 defines land-surface forecasting as future satellite prediction guided by meteorology (Requena-Mesa et al. 2021); early analysis showed that explicit weather conditioning affects both fore-

cast skill and simulated response (Diaconu et al. 2022). GreenEarthNet and Contextformer focus the task on vegetation dynamics and introduce a strong multimodal, high-resolution baseline (Benson et al. 2024). ConvLSTM, PredRNN, SimVP, and Earthformer represent recurrent-memory, convolutional, and transformer approaches commonly adapted to this task (Shi et al. 2015; Wang et al. 2017; Gao et al. 2022b,a). MCVD and VegeDiff instead model multiple plausible futures (Voleti, Jolicoeur-Martineau, and Pal 2022; Zhao et al. 2025). ViT-Koop advances compressed EO states with a linear Koopman operator for efficient forecasting (Shinohara 2025). TerraState instead places an exposed state in the forecast closure and tests, within the same checkpoint, whether that state is load-bearing and responds to the declared driver.

EO world models and forcing response. EO-WM frames EO forecasting as partially observed, weather-driven world modeling, decomposes climatology, anomalies, and accumulated stress, and evaluates extreme-summer and same-location cross-year response (Luo et al. 2026). VegSim is the closest architectural precedent: it infers a latent vegetation state, rolls it recurrently under future weather, and decodes NDVI quantiles for forecasting and scenario simulation (Iele et al. 2026). Cloud-aware observability forecasting uses a latent world model for a different target—whether and when the next usable acquisition will occur (Albughdadi 2026). Complementing output-level response tests, TerraState ties an exposed state to the forecast itself, cuts that contribution under a matched checkpoint, and changes the forcing while tracking both state and output.

Predictive states and latent dynamics. Predictive-state representations define state through predictions of future observables rather than an unobserved physical variable (Littman, Sutton, and Singh 2001). Modern world models learn compact dynamics for prediction and control (Ha and Schmidhuber 2018; Hafner et al. 2019, 2020), while joint-embedding architectures predict representations rather than raw pixels (Assran et al. 2023; Bardes et al. 2024). LatentTSF emphasizes state-space forecasting (Yang et al. 2026), and PLSM regularizes how controls change latent states (Saunum, Dayan, and Schulz 2024). Deep-OSG studies semigroup-aware variable-time operators for autonomous systems (Chen and Wu 2023); concurrent work tests identity, inverse, and composition relations induced by group-structured agent actions (Wang et al. 2026). Those algebraic settings differ from ordered, generally irreversible weather forcing; our optional path test therefore compares two partitions of the same observed forcing interval.

EO representation learning. Large-scale EO pretraining exploits seasons, multispectral time series, and cross-sensor views (Mañas et al. 2021; Cong et al. 2022; Wang et al. 2023; Fuller, Millard, and Green 2023; Guo et al. 2024); cloud-removal work separately models uncertainty in optical time series (Ebel et al. 2023). Such encoders can initialize the context operator, but a strong static representation does not by itself establish a forecast-bearing transition. Our diagnostics apply after the representation is placed on the forecasting path.

3 Method

Problem Setting and Predictive-State Criterion

Each example contains a cloud-obscured optical history $\mathcal{H}_t = \{(x_i, m_i, \tau_i)\}_{i=1}^C$, a future meteorological path $u_{t:t+H}$, static geographic context g , and future land-surface targets $y_{t+1:t+H}$. Here m_i marks valid pixels and τ_i records acquisition time; $u_{a:b}$ uses half-open indexing and contains steps $a, \dots, b-1$. The public task uses $C = 10$ five-day context observations and $H = 20$ forecast endpoints (Benson et al. 2024). Let $\tilde{\mathcal{C}}_t = (\mathcal{H}_t, u_{\leq t}^{\text{past}}, g)$ denote the retained context: cloud-masked EO history, past meteorology, and static geography, with future satellite frames, future target masks, and future weather suppressed. TerraState represents a query at horizon h as

$$\begin{aligned} (b_{1:H}, z_t) &= q_{\theta, \rho}(\tilde{\mathcal{C}}_t), \\ z_{t+h} &= T_\psi(z_t, u_{t:t+h}, g, h), \\ r_h &= O_\omega(z_{t+h}), \quad \hat{y}_{t+h} = b_h + r_h, \end{aligned} \quad (1)$$

where b_h is a context-only forecast prior and r_h is the contribution carried by the transitioned state. Future weather can reach the final forecast only through T_ψ ; b_h cannot change under a Q3 weather intervention. Here $q_{\theta, \rho}$ denotes the composite context/state operator: below, q_θ is its initialized forecasting backbone and P_ρ its state projector.

We call z_t a *predictive state* when it is inferred without future target observations, contributes to endpoint prediction, and supports queries under different horizons and forcing paths. The first two properties are tested by the forecast and

closure interventions; the third is probed by matched weather changes and, optionally, path composition.

Context-Only Spatial State

To expose a spatial state without adding a second forecasting path, the inference operator is initialized from a PVT v2/Contextformer forecaster (Wang et al. 2022; Benson et al. 2024). On $\tilde{\mathcal{C}}_t$ it embeds retained past meteorology and static channels together with the cloud-masked EO context, returns a horizon-indexed prior, and exposes the final context tokens before the forecast head. A spatial projector converts those tokens into

$$z_t = P_\rho(q_\theta(\tilde{\mathcal{C}}_t)_{\text{context end}}), \quad z_t \in \mathbb{R}^{N \times d}. \quad (2)$$

P_ρ is a LayerNorm–linear–GELU–linear–LayerNorm projector with widths $256 \rightarrow 512 \rightarrow 256$. N indexes spatial patches and d is the state width. For a 128×128 minicube and 4×4 patches, $(N, d) = (32^2, 256) = (1024, 256)$; the implementation flattens a batch to $[1024B, 256]$. Cloud masks determine which historical observations are usable. Both the prior $b_{1:H}$ and state z_t are computed from the same context-only pass. This shared origin lets Q2 remove the state contribution while preserving the history-conditioned prior; no second inference backbone or training teacher is needed at test time.

Shared Weather-Driven Transition

The transition should attribute driver response to one reusable operator rather than to separate horizon-specific heads. For a half-open interval $[a, b)$, a shared GRU summarizes the *ordered* meteorological segment. Its input contains eight standardized E-OBS variables, each aggregated by five-day mean, minimum, and maximum, for 24 channels. Missing values are mapped to zero in this standardized space. Static geography uses the three standardized elevation products averaged to the state patch grid and passed through a two-layer MLP; elapsed time uses a fixed sinusoidal embedding. The same static fields are available to q_θ in the retained context and are separately encoded here for the transition:

$$e_{a:b} = E_u(u_{a:b}), \quad c_{a:b} = F(e_{a:b}, E_g(g), E_h(b-a)). \quad (3)$$

The three codes are fused by a two-layer MLP. One controlled residual transition—LayerNorm followed by a three-layer GELU MLP with a residual connection—is reused at every horizon and on every path:

$$T_\psi(z_a, u_{a:b}, g, b-a) = z_a + \Delta_\psi(\text{LN}(z_a), c_{a:b}). \quad (4)$$

Sharing ψ across horizons makes its weather response testable in Q3. Interval encoding preserves weather order and is recomputed for each subinterval, which also permits the optional direct/composed query in Q4.

Forecast Closure and Recoverability

The closure places the transitioned state on the prediction path while retaining an exact same-checkpoint removal test.

It decodes each advanced spatial token and unpatchifies the result into an endpoint residual:

$$r_h = O_\omega(z_{t+h}), \quad \hat{y}_{t+h}^{(s)} = b_h + s r_h, \quad s \in \{0, 1\}. \quad (5)$$

Normal inference fixes $s = 1$; s is not a learned gate. The diagnostic setting $s = 0$ exactly recovers the context-only prior while leaving the checkpoint and sample population unchanged. TerraState therefore claims only that the state carries a measurable forecast increment beyond b_h , not that every accurate component of the forecast passes through z_{t+h} . The internal context-only prior is also distinct from the separately trained matched-backbone comparator used in Q1.

Optional Direct and Composed Evaluation Queries

To test whether the shared transition can be reused over subintervals, for $h = h_1 + h_2$ TerraState exposes two paths from the same state under the same ordered forcing:

$$z_h^{\text{dir}} = T_\psi(z_t, u_{t:t+h}, g, h), \quad (6)$$

$$z_h^{\text{cmp}} = T_\psi(T_\psi(z_t, u_{t:t+h_1}, g, h_1), u_{t+h_1:t+h}, g, h_2). \quad (7)$$

The direct path applies T_ψ once over the full interval; the composed path reuses it over an evaluation partition. Both states are closed to forecasts $\hat{y}_h^{\text{dir}}$ and $\hat{y}_h^{\text{cmp}}$ and compared with the same true endpoint y_h . These queries are never optimized: after training and checkpoint selection, optional Q4 applies them to a predeclared evaluation set $\mathcal{P}_{Q4}$. Agreement therefore measures post-training behavior rather than a composition objective seen during optimization.

Training Objective and Curriculum

Training proceeds in three phases after *forecast initialization* warm-starts the PVT/Contextformer context operator and compatible TerraState modules from a forecasting checkpoint. We train for 40 epochs and 14,880 optimizer updates (372 updates per epoch). In phase 1, steps $[0, 2976)$, q_θ is frozen and λ_s rises linearly from 0 to 0.02. In phase 2, steps $[2976, 11904)$, q_θ remains frozen and $\lambda_s = 0.02$; a checkpoint is forced at the 80% boundary. In phase 3, steps $[11904, 14880)$, only `core.blocks.2`, the final Transformer block of q_θ , is unfrozen and $\lambda_s = 0.01$. All earlier blocks remain frozen.

Optimization uses AdamW with $\beta = (0.9, 0.999)$, zero weight decay, and global batch 64 (eight GPUs, eight samples per GPU, no gradient accumulation). The non- q branch has base learning rate 3×10^{-5} ; the q parameter group uses 0.033 times that rate, 9.9×10^{-7} , and receives updates only in phase 3. A shared multiplier warms linearly from zero for 300 optimizer steps and then follows cosine decay; gradients are clipped at norm 1.0. Validation-only selection freezes one checkpoint for Q1–Q3, optional Q4, and the one-shot temporal-shift evaluation.

Let $\mathcal{P}$ be the vegetation pixels with a valid prediction and $v_{p,h} \in \{0, 1\}$ indicate that pixel p is clear at horizon h . The forecast loss first averages over the visible horizons of each pixel and then averages over pixels:

$$\mathcal{L}_{\text{GT}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \frac{\sum_{h=1}^H v_{p,h} (\hat{y}_{p,h} - y_{p,h})^2}{\sum_{h=1}^H v_{p,h} + \epsilon}. \quad (8)$$

A strong full24 forecasting teacher is frozen throughout training and discarded at inference. Its prediction $\hat{y}_h^{\text{teach}}$ supplies the single knowledge-distillation term

$$\mathcal{L}_{\text{KD}} = \frac{\sum_{p,h} v_{p,h} (\hat{y}_{p,h} - \text{sg}(\hat{y}_{p,h}^{\text{teach}}))^2}{\sum_{p,h} v_{p,h} + \epsilon}. \quad (9)$$

Future-state supervision is generated before optimization by a frozen copy of the initialized q_θ and P_ρ . For valid terminal patches $i \in \{1, \dots, 1024B\}$ with mask m_i , it is

$$z_{t+20,i}^* = \text{sg}[P_{\text{frozen}}(q_{\text{frozen}}(o_{\leq t+20}))_i],$$

$$\ell_i^{\text{state}} = 1 - \cos(\text{LN}(z_{t+20,i}), \text{LN}(z_{t+20,i}^*)), \quad (10)$$

$$\mathcal{L}_{\text{future-state}} = \frac{\sum_i m_i \ell_i^{\text{state}}}{\sum_i m_i + \epsilon}.$$

The target input contains the observed future EO sequence, has future weather zeroed, and supplies only the terminal $h = 20$ target. A terminal 4×4 patch is valid only if it is fully cloud-free and contains at least one vegetation pixel. Each cube’s $[1024, 256]$ target is cached in FP16 together with its Boolean mask, relative source path, and provenance checksums. Targets are computed once into separate train and validation caches, and training consumes only those cached tensors. Neither the target encoder nor the forecasting teacher is part of the inference model. The only nonzero training objective is

$$\mathcal{L} = \mathcal{L}_{\text{GT}} + 0.5 \mathcal{L}_{\text{KD}} + \lambda_s \mathcal{L}_{\text{future-state}},$$

$$\lambda_s = \begin{cases} \text{linear } 0 \rightarrow 0.02, & 0\% \text{--} 20\%, \\ 0.02, & 20\% \text{--} 80\%, \\ 0.01, & 80\% \text{--} 100\%. \end{cases} \quad (11)$$

Q2–Q4 cuts, weather interventions, path comparisons, and non-collapse statistics are post-training measurements and receive no gradient. In particular, future-state supervision is a training signal rather than evidence that Q2 has passed; load-bearing state use must still be established by endpoint interventions.

Matched State Diagnostics

Load-bearing. We report full, closure-cut, and transition-to-identity forecasts on identical samples. The closure cut sets the state-carried increment to zero ($s = 0$ in Equation 5) and therefore recovers b_h without altering the inferred state. The transition cut replaces T_ψ by the identity while keeping the closure, decoder, and all other inputs fixed. A state is credited as load-bearing only if these cuts remove a measurable paired full-versus-reference gain and change the endpoint output. Because $T \rightarrow I$ can feed O_ω a state outside its trained input distribution, the exact closure cut is primary and the transition cut is supporting mechanism evidence.

Driver sensitivity. Matched future weather is compared with (i) a season- and geography-matched, distinct donor sequence and (ii) the normalized-mean control obtained by zeroing weather after the fixed centering transform. The donor manifest, matching keys, and normalization record are set from validation before test evaluation. We keep history, geography, reference forecast, masks, and checkpoint fixed;

change only the declared transition input; and report paired changes in transitioned state, endpoint output, and forecast score. An unmatched within-batch shuffle is used only for engineering smoke tests.

Optional post-training path consistency with endpoint guards. After freezing the checkpoint, for each predeclared Q4 partition p we report direct and composed endpoint errors E_p^{dir} and E_p^{cmp} separately. The metric and thresholds are fixed on validation data. A partition qualifies only when both endpoints satisfy the absolute guard and the composed path meets a predeclared non-inferiority margin:

$$G_p = \mathbb{I} \left[\begin{array}{l} \max(E_p^{\text{dir}}, E_p^{\text{cmp}}) \leq \tau_{\text{abs}} \\ \wedge E_p^{\text{cmp}} \leq (1 + \epsilon_{\text{NI}}) E_p^{\text{dir}} \end{array} \right]. \quad (12)$$

For qualified partitions, we report

$$\begin{aligned} \delta_z &= \text{mean} |z_h^{\text{dir}} - z_h^{\text{cmp}}|, \\ \delta_y &= \ell_{\mathcal{V}_h}(\hat{y}_h^{\text{dir}}, \hat{y}_h^{\text{cmp}}), \\ A_{\text{comp}} &= \delta_y^{\text{broken}} - \delta_y. \end{aligned} \quad (13)$$

The real second leg consumes $u_{t+h_1:t+h_1+h_2}$. The broken control instead feeds that leg the equal-length prefix $u_{t:t+h_2}$, while retaining its elapsed-time embedding h_2 ; the first leg, direct path, geography, prior, closure, and target are unchanged. The half-open windows neither duplicate nor drop a boundary step on the real path. Thus A_{comp} asks whether the real path agrees better than a deliberately misaligned path. Path agreement is withheld from positive interpretation whenever $G_p = 0$, and Q4 cannot compensate for a failed Q2.

Non-degeneracy. Within each minicube, we compute mean absolute movement $M_h = \text{mean}_{n,c} |z_{t+h,n,c} - z_{t,n,c}|$, average the per-channel standard deviation across patch tokens to obtain S_h , and form the patch-token covariance after channel-wise centering. Its effective rank is

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2 + \epsilon}, \quad (14)$$

where λ_i are eigenvalues of the patch-token covariance. We also report S_h/S_t and $r_{\text{eff},h}/r_{\text{eff},t}$. These quantities guard against identity dynamics, constant states, and deceptively low path gaps. They are computed at horizons $\{1, 5, 10, 20\}$ on the same minicubes used by the path test and then averaged over minicubes; across-minicube movement variance is retained as an additional audit field.

4 Experiments

Questions and Protocol

Following the claim structure, the experiments answer four predeclared questions:

1. **Q1—Forecast skill:** Is the selected TerraState checkpoint competitive under temporal distribution shift and against the matched backbone?
2. **Q2—Load-bearing:** Does cutting the state-carried contribution or its transition remove a measurable forecasting contribution?

3. **Q3—Driver sensitivity:** Does matched weather improve prediction relative to matched-donor and normalized-mean forcing?
4. **Q4—Optional composition and non-collapse:** After training, are direct/composed queries individually accurate, mutually compatible, and non-degenerate?

Data and validation-only selection. We use GreenEarthNet (Benson et al. 2024). Each minicube contains 30 five-day Sentinel-2 composites at 128×128 pixels and 20 m ground sampling, aligned meteorology, a cloud/quality mask, and elevation. The first 10 composites form the context and the following 20 define the target horizon. Architecture, loss, decision thresholds, the donor rule, and checkpoint selection use training and validation data only. The donor rule specifies its season, geography, day-of-year window, reuse cap, and divergence-quantile level before temporal-shift evaluation. The per-track assignment and its absolute divergence floor are then computed deterministically from input-side time, location, and provided future weather before model scoring; they never use targets, predictions, or errors. This transductive use of declared test covariates is recorded and hashed. After the remaining choices are fixed, the selected checkpoint is evaluated once on the available `ood-t_chopped` temporal-shift track with 1,904 minicubes; this track is never used for checkpoint selection, early stopping, or performance-threshold tuning. We do not describe this subset as the complete GreenEarthNet protocol because the local evaluation bundle does not contain every official split.

Targets and metrics. The primary target is NDVI over pixels that are both valid and in the declared vegetation land-cover range. We report R^2 , RMSE, NSE, absolute bias, out-performance, and RMSE25 using one fixed scorer and one aggregation convention. NSE is Nash–Sutcliffe efficiency; RMSE25 is RMSE over the first five five-day endpoints; and out-performance is the fraction of aligned minicube–land-cover groups that beat the fixed climatology reference on at least three of four prescribed criteria. EarthNetScore, when used for an optional EarthNet2021 analysis, is reported in a separate table and never mixed with GreenEarthNet metrics.

Comparisons. Published baselines are transcribed from the GreenEarthNet/Contextformer report (Benson et al. 2024) and labeled Reported. They provide historical context only. The exact local panel contains the already available matched forecasting backbone and the selected TerraState checkpoint; no local reproduction of ConvLSTM, PredRNN, SimVP, Earthformer, or Contextformer is required. Every paired claim is restricted to these local rows sharing target files, masks, horizons, scorer, and aggregation code. The matched backbone uses the same forecasting lineage and declared local protocol but omits TerraState’s state projector, transition, and closure; any remaining budget or teacher-access difference is reported as a confound. We never claim a strict ranking across the published and local panels.

Implementation and statistics. TerraState uses a context-only PVT/Contextformer prior, a projected spatial state, the shared transition in Equation 4, and the residual closure in

Equation 5. It contains 7,180,896 unique `nn.Parameter` scalars, including the frozen q backbone but excluding buffers (including the fixed closure coefficient), the frozen KD teacher, the offline future-target encoder, and its cache. Of these, 1,120,336 are trainable in phases 1–2 and 1,910,096 in phase 3. Its per-minicube state shape is $[1024, 256]$ (runtime $[1024B, 256]$). The 40-epoch, 14,880-update AdamW configuration, learning-rate groups, batch 64, and three-phase schedule are specified above; the checkpoint is selected on **[FROZEN FULL-VALIDATION CRITERION]**. The primary Q1–Q3 report and optional Q4 use this one checkpoint and paired minicube-level bootstrap intervals. If **[REPLICATION SEEDS]** independent replications are completed, they are secondary robustness evidence, not an average substituted for the checkpoint-level state claim; without them, Q1 is labeled a single-checkpoint comparison rather than a general training-stability claim. Before the one-shot test, the reproducibility record stores hashes for the student initialization, training-only teacher, frozen future-target encoder/projector and target-cache manifest, final checkpoint, serialized model configuration, three-term objective configuration, validation selection record, data manifest, evaluator, mask/aggregation protocol, donor manifest, endpoint guard, and optional Q4 evaluation partitions.

Q1: Forecast Skill under Temporal Shift

Table 1 separates published context from the exact local matched comparison. The final result paragraph will make only one of three evidence-supported claims: (i) improvement over the matched backbone with a paired confidence interval, (ii) validation-predeclared non-inferiority together with an additional state capability, or (iii) a negative result that triggers claim reduction. Without a predeclared margin, option (ii) is unavailable. The paragraph will not mix the published Contextformer aggregation with a locally computed score or describe a single-seed pilot as state of the art.

Q2–Q3: Load-Bearing State and Weather Response

Table 2 separates absolute forecast accuracy from the incremental contribution of the state branch. Q2 compares the full checkpoint with its recoverable closure cut and a transition-to-identity cut; the matched backbone remains the Q1 reference. Q3 changes only the declared transition input while holding history, geography, closure, masks, sample population, and checkpoint fixed.

Optional Post-Training Q4: Composition and Anti-Collapse

Q4 is a post-training extension and may be moved to the supplementary document. It uses the validation-frozen partition set

$$\mathcal{P}_{Q4} = \{(3, 7), (6, 4), (4, 11), (8, 12), (2, 18)\}$$

only after the unique checkpoint has been selected and frozen; these partitions do not define training examples or losses. Every partition is evaluated on the same endpoint population. A row receives a positive path-consistency interpretation only if its direct and composed endpoints pass

Equation 12, its real output gap is compared with the asymmetric broken-path control, and the state-retention measurements do not indicate collapse. Q4 neither changes the checkpoint nor participates in model selection.

Same-Checkpoint Diagnostics and Optional External Analyses

The required mechanism evidence uses the selected checkpoint without retraining: Q2 applies the closure and $T \rightarrow I$ cuts, while Q3 changes only the future-weather input. We do not require retrained removals of the projector, transition, conditioning encoders, KD term, or future-state anchor for the main claims. If later conducted, such studies will be labeled supplementary rather than substituted for Q1–Q3. Independent EO-WM response protocols (Luo et al. 2026) are likewise optional external analyses, not part of the required evidence.

Qualitative analysis. A qualitative panel is included only after real arrays are available. It uses one validation-selected tile and shows target NDVI, full/closure-cut forecasts, matched/donor change, and direct/composed change on shared scales with valid-pixel coverage. The identifier is frozen before test scoring and is never chosen from test errors or saliency.

5 Limitations and Scope

TerraState learns a future-predictive representation constrained by satellite history and supplied meteorology; it does not recover a complete physical land-surface state. Future weather in the benchmark is an oracle exogenous input, so operational performance with forecast rather than observed weather may be lower.

Matched-versus-donor sensitivity measures conditional response, not causal identification or counterfactual correctness. Direct/composed agreement is an empirical finite-horizon diagnostic, not a proof of physical validity, associativity, or a semigroup law. The residual closure supports only the claim that the state-mediated branch carries a measurable forecast increment, not that every accurate output component or the history content of z_t is indispensable. Because Q4 is not optimized, failure of its optional post-training test limits any composition claim but does not overwrite the separately reported Q1–Q3 findings.

The contract is a matched verification protocol for the selected TerraState checkpoint, not an independent benchmark, and one temporal-shift track cannot establish cross-dataset generality. Optical targets also remain affected by cloud screening and unobserved variables such as soil moisture, irrigation, and vegetation type.

6 Conclusion

TerraState makes a world-state claim falsifiable: one accurate checkpoint must expose a load-bearing state whose shared transition uses the declared weather driver. Q1–Q3 test these core claims; optional Q4 probes guarded composition.

Method	Source	$R^2 \uparrow$	RMSE $\downarrow$	NSE $\uparrow$	$ \text{Bias} \downarrow$	RMSE25 $\downarrow$	Outperf. $\uparrow$
<i>A. Published context (reported values)</i>							
Persistence	Reported	0.00	0.23	-1.28	0.17	0.09	21.8%
Previous year	Reported	0.56	0.20	-0.40	0.14	0.18	19.3%
Climatology	Reported	0.58	0.18	-0.34	0.13	0.16	n.a.
ConvLSTM	Reported	0.58 ± 0.01	0.16 ± 0.00	-0.13 ± 0.02	0.11 ± 0.00	0.11 ± 0.00	$53.1 \pm 1.2\%$
Earthformer	Reported	0.52	0.16	-0.13	0.10	0.09	56.5%
PredRNN	Reported	0.62 ± 0.00	0.15 ± 0.00	0.03 ± 0.00	0.10 ± 0.00	0.10 ± 0.00	$64.7 \pm 1.2\%$
SimVP	Reported	0.60 ± 0.00	0.15 ± 0.00	0.03 ± 0.01	0.09 ± 0.00	0.10 ± 0.00	$64.1 \pm 1.0\%$
Contextformer	Reported	0.62 ± 0.00	0.14 ± 0.00	0.09 ± 0.01	0.09 ± 0.00	0.08 ± 0.00	$66.8 \pm 0.3\%$
<i>B. Exact local protocol (paired comparison only)</i>							
Matched backbone	Local	TBD	TBD	TBD	TBD	TBD	TBD
TerraState	Local	TBD	TBD	TBD	TBD	TBD	TBD

Table 1: Published forecast context and the local temporal-shift comparison. Panel A transcribes reported values from Table 2 of Benson et al. (2024); reported standard deviations use three seeds where available. Panel B contains only the matched backbone and TerraState under one local manifest and evaluator. Different protocols preclude strict cross-panel ranking. TerraState’s Q1–Q3 and optional Q4 results refer to the same frozen checkpoint; local confidence intervals and paired win/tie/loss counts accompany the machine-readable package.

Test	Variant	$R^2 \uparrow$	RMSE $\downarrow$	$\Delta_{\text{loss}} R^2$	$\Delta_{\text{rise}} \text{RMSE}$	State $ \Delta $	Output $ \Delta $
Q2	Full TerraState	TBD	TBD	0	0	0	0
Q2	Closure cut	TBD	TBD	TBD	TBD	–	TBD
Q2	$T \rightarrow I$ cut	TBD	TBD	TBD	TBD	TBD	TBD
Q3	Matched weather	TBD	TBD	0	0	0	0
Q3	Normalized-mean weather	TBD	TBD	TBD	TBD	TBD	TBD
Q3	Season/geography donor	TBD	TBD	TBD	TBD	TBD	TBD

Table 2: Matched Q2–Q3 interventions on identical samples. Q2 deltas are full-minus-variant for R^2 and variant-minus-full for RMSE; Q3 uses matched weather as reference, so positive means lost skill. State/output columns are mean absolute token/valid-pixel NDVI changes, and the Q3 context prior must be invariant. Because the closure cut equals that prior, we report its absolute paired effect rather than an algebraic “gain removed fraction.” Per-horizon values and intervals accompany the table.

<i>A. Endpoint accuracy and path controls by partition</i>							
Partition	E^{dir}	E^{cmp}	Joint guard	δ_z	δ_y	δ_y^{broken}	A_{comp}
3+7	TBD	TBD	TBD	TBD	TBD	TBD	TBD
6+4	TBD	TBD	TBD	TBD	TBD	TBD	TBD
4+11	TBD	TBD	TBD	TBD	TBD	TBD	TBD
8+12	TBD	TBD	TBD	TBD	TBD	TBD	TBD
2+18	TBD	TBD	TBD	TBD	TBD	TBD	TBD
<i>B. Anti-collapse summaries by direct-query horizon</i>							
Horizon	M_h	S_h/S_t	$r_{\text{eff},h}/r_{\text{eff},t}$	Validation-frozen degeneracy check			
1	TBD	TBD	TBD	TBD			
5	TBD	TBD	TBD	TBD			
10	TBD	TBD	TBD	TBD			
20	TBD	TBD	TBD	TBD			

Table 3: Optional post-training Q4 separates partition-level path evidence (A) from horizon-level anti-collapse evidence (B). No Q4 quantity is a training loss or selection criterion. Panel A reports endpoint MSE, the joint guard, state/output gaps, the broken-path gap, and $A_{\text{comp}} = \delta_y^{\text{broken}} - \delta_y$. Panel B reports movement, standard-deviation retention, effective-rank retention, and the validation-frozen degeneracy check. Positive interpretation requires a passing joint guard and cannot compensate for failed Q2.

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